

PROJECT SYNOPSIS

Final Year Minor Project Synopsis — B.E. Computer Science & Engineering

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Program / Branch	B.E. (Computer Science & Engineering)	Semester	7th
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1. Title

Real-Time Sign Language Translator: A computer vision application using OpenCV and MediaPipe to detect hand gestures via webcam and translate them into text and speech in real time.

2. Brief Introduction

Background

Sign language is the primary mode of communication for millions of people who are deaf or hard of hearing, yet the vast majority of hearing individuals have no working knowledge of it, creating a persistent communication barrier in everyday settings. Advances in computer vision and lightweight machine learning have made it possible to bridge this gap without specialized hardware such as gloves or depth sensors. Google's MediaPipe framework offers pretrained, real-time hand-landmark detection that runs efficiently on consumer hardware, while OpenCV provides the tooling to capture and process live video streams. Together, these libraries allow a standard webcam to be turned into a gesture-recognition sensor, and a lightweight classifier layered on top can map hand poses to letters or words, which can then be rendered as on-screen text and read aloud through text-to-speech. This project builds on that pipeline to create a practical, real-time translation tool.

Scope and Assumptions

The project targets recognition of a defined static gesture set (e.g., the alphabet and a small vocabulary of common words) captured via a single standard webcam under reasonably controlled lighting, using one hand or two hands in frame. Continuous, fluid sign language with complex grammar, facial-expression-based grammar markers, and multi-signer scenarios are explicitly out of scope, as they require substantially larger datasets and sequence models beyond a single-semester project. Where a design decision is genuinely complex (for example, distinguishing visually similar gestures under motion blur), the project will document the limitation and, where feasible, apply simplified heuristics such as temporal smoothing rather than attempting full linguistic-level sign recognition.

3. Aim and Objectives

Aim

To design and develop a real-time computer vision application that detects hand gestures via webcam using OpenCV and MediaPipe, classifies them into corresponding alphabets/words, and translates them into readable text and audible speech, thereby demonstrating a practical application of hand-landmark detection, feature-based gesture classification, and real-time human-computer interaction.

Objectives

- To implement real-time hand detection and extraction of 21-point landmark coordinates per hand from a live webcam feed.
- To design a normalized, scale- and translation-invariant feature representation from raw landmark coordinates.
- To collect/curate a gesture dataset and train a lightweight classification model mapping features to sign classes.
- To implement temporal smoothing and a sequence buffer that assembles individual predictions into words and sentences.
- To integrate a text-to-speech module that converts the recognized text output into audible speech in real time.
- To evaluate the system's correctness and performance by testing detection accuracy, classification accuracy, and end-to-end latency.

Module-wise Description

#	Module	Description
1	Video Capture & Preprocessing	Captures live webcam frames via OpenCV, resizes/normalizes them for model input.
2	Hand Landmark Detection	Uses MediaPipe Hands to detect and track 21 hand landmarks per hand in real time, extracting normalized coordinates.
3	Feature Extraction & Preprocessing	Converts raw landmark coordinates into a normalized, translation- and scale-invariant feature vector suitable for classification.
4	Gesture Classification Engine	A trained ML/DL model that maps the feature vector to a predicted alphabet/word/sign class.
5	Temporal Smoothing & Sequence Buffer	Maintains a rolling buffer of recent predictions to filter jitter, debounce repeated frames, and assemble letters into words/sentences.
6	Text-to-Speech (TTS) Module	Converts the recognized, smoothed text output into audible speech using an offline/online TTS engine.
7	User Interface	A live application window that overlays hand landmarks, displays the translated text stream, and triggers speech playback.

4. Tools to be Used

Category	Tool / Technology	Purpose
Programming Language	Python	Primary implementation language for all core modules and application logic.
Computer Vision	OpenCV	Captures webcam frames, handles image preprocessing, and renders the live annotated video feed.
Hand Tracking	MediaPipe	Provides pretrained, real-time hand landmark detection (21 keypoints per hand).
Machine Learning	scikit-learn / TensorFlow-Lite	Trains and runs the gesture classification model on extracted landmark features.
Text-to-Speech	pyttsx3	Converts recognized text output into audible speech in real time.

